

UNIVERSITY OF CAPE TOWN
DEPARTMENT OF STATISTICAL SCIENCES - STA2020S: APPLIED STATISTICS
CLASS TEST 1 – 5 September 2025

Time: 90
MINUTES

Max Marks: 50

Internal examiners: Mrs. D.M. van
Wyk-de Ridder
Ms. A. Cloete

Instructions: Answer ALL questions.
Marks are awarded for intermediate calculations.
Answer all MCQ questions *IN PENCIL* on the *MCQ sheet* provided.
Answer each written question *IN PEN* on a *NEW PAGE* in your answer book.
ROUND ALL ANSWERS TO 2 DECIMAL PLACES UNLESS OTHERWISE SPECIFIED.
DO NOT ROUND INTERMEDIARY CALCULATIONS.

SECTION A: MCQ QUESTIONS

[10]

Use the following information to answer questions 1. and 2. of SECTION A:

A correlation analysis is done between a continuous dependent variable, y , and a continuous independent variable, x , to determine whether the correlation coefficient between these two variables is significant or not. It is known that the test statistic for this test is equal to -1.6132 , and that $SS_x = \sum_{i=1}^{50} (x_i - \bar{x})^2 = 12\,976.59$, $SS_y = \sum_{i=1}^{50} (y_i - \bar{y})^2 = 6\,031.094$, and $SS_{xy} = \sum_{i=1}^{50} (x_i - \bar{x})(y_i - \bar{y}) = -63\,441.65$.

1. Choose the **correct** answer about this correlation analysis: (2)
 - A. Under the null hypothesis, the test statistic follows a t distribution with 49 degrees of freedom.
 - B. The null hypothesis is that the population correlation coefficient is equal to 0.
 - C. Under the null hypothesis, the test statistic follows a standard normal distribution.
 - D. The alternative hypothesis is that the sample correlation coefficient is not equal to 0.
 - E. Correlation is a measure of x 's strength to predict y .
2. Choose the **incorrect** answer about this correlation analysis: (2)
 - A. The sample correlation coefficient indicates a negative linear relationship between x and y .
 - B. At a 5% level of significance, we would conclude that there is no significant linear relationship between x and y .
 - C. At a 10% level of significance, we would conclude that there is a significant linear relationship between x and y .
 - D. The size of the sample correlation coefficient is indicative of a weak linear relationship between x and y .
 - E. The correct code for conducting this correlation analysis in R would be: `cor.test(x,y)`.

3. A logistic regression model was fit for $n=36$ observations of a binary dependent variable (1 = event occurs, 0 = event does not occur) and two continuous explanatory variables, i.e., $\text{logit}(p_i) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$. It is known that the lower bound of the 95% confidence interval for β_2 is -1.14 and $se(\hat{\beta}_2) = 0.11$. Choose the **correct** answer, given this information. (2)
- A. $\hat{\beta}_2 = -0.924$, rounded to three decimal places.
 - B. $\hat{\beta}_2 = -1.363$, rounded to three decimal places.
 - C. $\hat{\beta}_2 = -1.356$, rounded to three decimal places.
 - D. $\hat{\beta}_2 = -0.916$, rounded to three decimal places.
 - E. $\hat{\beta}_2 = -0.959$, rounded to three decimal places.
4. Two multiple linear regression models are fit to the same dataset, which contains a continuous dependent variable and two continuous independent variables. The following table contains important information about the two models:

	Model A: $y \sim \beta_0 + \beta_1 x_1 + \beta_2 x_2$	Model B: $y \sim \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$
RSE	21.56	16.22
AIC	306.23	308.12
R^2	0.87	0.89

- Choose the **incorrect** answer about the two models: (2)
- A. Given the higher RSE value, it is plausible to deduce that the accuracy of the predictions of Model A is better than that of Model B.
 - B. A possible reason for observing a higher AIC value for Model B could be that we include an additional parameter to be estimated in this model.
 - C. The coefficient of determination indicates that the independent variables in both models explain at least 85% of the variation in the dependent variable.
 - D. A lower RSE value indicates a lower sum of squared errors.
 - E. The tabled results are implausible because Model B has a lower RSE than Model A, so it must also have a lower AIC.
5. A psychologist wants to compare average stress levels between three relaxation techniques: yoga, meditation, and breathing exercises. She recruits 30 students and assigns 10 to each technique. The yoga group consists of friends from the same sports club, the meditation group are all classmates from one seminar, and the breathing group are strangers recruited individually.

Which assumption of the one-way ANOVA is most likely violated, and why? (2)

- A. Equal variance — because stress levels may vary more in some groups than others.
- B. Normality — because stress scores may not follow a normal distribution.
- C. Independence — because students within the same group may influence each other's stress levels.
- D. Random sampling — because the researcher did not select participants randomly from the population.
- E. Additivity — because the effects of the techniques may interact with one another.

SECTION B: REGRESSION QUESTIONS

[30]

QUESTION 1

[8]

Shoes by Randy is a chain of stores that specialises in stocking trainers and sneakers. The executive committee wants to understand what drives the monthly sales revenue of retail outlets. The committee tasks their in-house data analyst to review the data from the past 50 months.

The following variables are available:

- Monthly Sales Revenue (*SalesRev*) – measured in ten thousands of Rands (R10 000).
- Advertising Spend (*AdSpend*) – measured in thousands of Rands (R1 000).

The analyst fits a model to the data in *R Studio* and generates (partial) output below. It is further known that the sum of squared error terms is given as 4314730.

```
Call:
lm(formula = SalesRev ~ AdSpend, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-624.59 -176.44  -17.31   241.23   822.53

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 1042.4138    112.5204   9.264 2.91e-12 ***
AdSpend       2.5464     0.5827   2.531      0.015
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error:
Multiple R-squared:  0.2846,    Adjusted R-squared:  0.2697
F-statistic: 19.09 on 1 and 48 DF,  p-value: 6.632e-05
```

- 1.1 The analyst tasks you with conducting a hypothesis test to assess the significance of the slope estimate. The test statistic has been calculated, but the analyst is unsure about the p-value. Determine the p-value as accurately as possible and inform the analyst of the conclusions for the test. Use a significance level of 5%. (2)

$(2 \times 0.005 = 0.01 < p\text{-value} < 2 \times 0.01 = 0.02)$ (1 mark) < 0.05 , therefore, the null hypothesis should be rejected. The slope coefficient is significant, which indicates that there is a significant linear relationship between sales revenue and advertising spend (1 mark).

Notes:

Give 0.5 marks for each endpoint of the interval for the p-value (be strict on this) **OR** give 0.5 marks if they state that the p-value is less than the significance level at 5% **OR** give 1 mark if they interpolate the endpoints and state p-value = 0.015.
Give 0.5 marks if they only state null should be rejected, **and** slope coefficient is significant.

No mark for stating that the null hypothesis should be rejected without indication of what this implies for the test (explanation/discussion/conclusion).
 No mark if the critical value approach was used, instead of the requested p-value approach.

- 1.2 Interpret the value of the coefficient of determination within the context of the fitted model. What does this value imply about the model fit?
 (2)

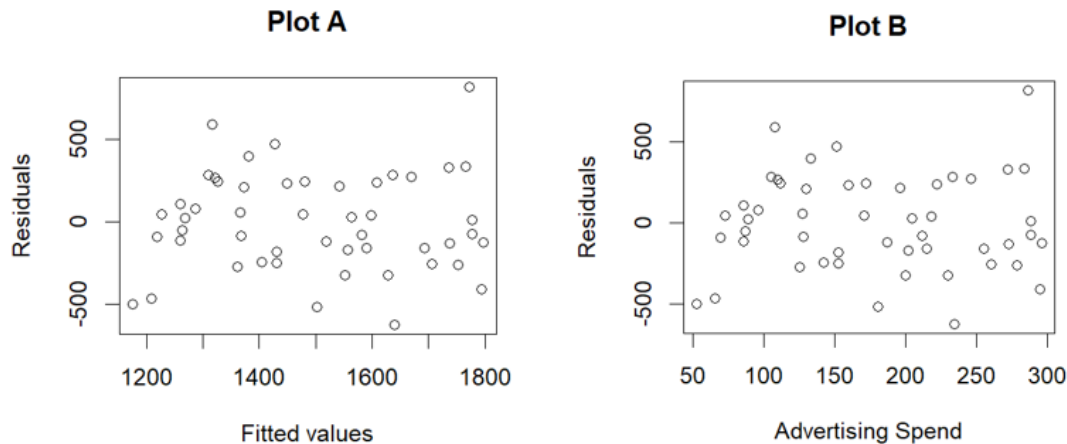
28.46% of variation in sales revenue is explained by the advertising spend (1 mark). This value is low and indicates a poor model fit (1 mark).

Notes:

Only award 0.5/1 for the first mark if the answer is correct, but non-specific (not referring to the model variables explicitly).

- 1.3 The analyst is confused about the results he is obtaining from investigating the underlying model assumptions. The following two plots, Plot A and Plot B, were generated in R. Explain the difference between the two plots by indicating which model assumption each is testing. State whether the model assumptions under consideration are being met in each case.
 (3)

Plot A: Assesses the assumption of constant variance for the errors (1 mark). Since there is no pattern, just random scatter, the assumption is met (0.5 marks).
 Plot B: Assesses the assumption of the independence of the errors (1 mark). Since there is no pattern, just random scatter, the assumption is met (0.5 marks).



- 1.4 Suppose it is known that $\sum_{i=1}^{50} (AdSpend_i - \overline{AdSpend})^2 = 264702.5$. Calculate the value of the residual standard error.
 (1)

$$RSE = se(\beta_1) \sqrt{SS_x} = 0.5827 \sqrt{264702.5} = 299.79. \text{ (1 mark)}$$

QUESTION 2

[13]

Shoes by Randy decides to include additional variables in their quest towards understanding the drivers of their monthly sales revenue. The data analyst is tasked with adapting the current model to include the full set of variables listed below:

- Monthly Sales Revenue (*SalesRev*) – measured in ten thousands of Rands (R10 000).
- Advertising Spend (*AdSpend*) – measured in thousands of Rands (R1000).
- Number of Active Customers (*Customers*) – in hundreds of customers.
- Regions of town where the stores are located (*Region*) – North, South, East, West.
- Store Size (*StoreSize*) – in square meters (m²).
- Average Price of Key Product (*AvgPrice*) – in Rands per unit.
- Store Type (*StoreType*) – Franchise, Independent.

The analyst fits a model to the data in *R Studio* and generates the following (partial) output:

```
Call:
lm(formula = SalesRev ~ AdSpend + Customers + Region + StoreSize +
    AvgPrice * StoreType)

Residuals:
    Min       1Q   Median       3Q      Max
-138.79  -66.98   -6.43   55.88  269.38

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   282.65619   151.50964    1.866   0.0694 .
AdSpend         3.13544    0.18889   16.600 < 2e-16 ***
Customers       2.05248    0.12970   15.825 < 2e-16 ***
RegionNorth    72.67649    36.85081    1.972   0.0555 .
RegionSouth   -85.71487    36.30005   -2.361   0.0232 *
RegionWest    -77.91474    45.43525   -1.715   0.0941 .
StoreSize       0.43694    0.05611    7.788 1.54e-09 ***
AvgPrice      -1.01361    1.12199   -0.903   0.3717
StoreTypeIndependent 46.79680   226.81419    0.206   0.8376
AvgPrice:StoreTypeIndependent -1.98533    1.91529   -1.037   0.3062
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 93.39 on 40 degrees of freedom
Multiple R-squared:  0.9422,    Adjusted R-squared:  0.9291
F-statistic:                p-value: < 2.2e-16
```

2.1 The analyst is interested in assessing the overall model significance. It is known that $SS_{resid} = 348866.4$ and $MS_{reg} = 631358.6$.

- a) Write down the null and alternative hypotheses for the test that the analyst needs to conduct. (1)

$H_0: \beta_1 = \beta_2 = \dots \beta_9 = 0$ (0.5 mark)

H_1 : At least one of the β_j 's are not equal to 0. (0.5 mark)

Notes:

Only award 0.5 if students do not specify β_9 , but a vague β_p .

- b) Calculate the relevant test statistic. (1)

$$F_{stat} = \frac{MS_{reg}}{SS_{resid}/(n-p-1)} = \frac{631358.6}{348866.4/40} = 72.39 \text{ (1 mark)}$$

Notes:

Do not award any marks if the wrong (switched) formula or wrong degrees of freedom is used. If the values were substituted correctly, but the final answer is calculated incorrectly, you may award 0.5 marks. You may also award 0.5 marks if the correct formula is used, but the degrees of freedom for SSresid is wrong.

- c) Using the p-value approach, with a significance level of 5%, what should the analyst conclude about overall model significance? (1)

The p-value (from the R output) < 0.05, therefore the conclusion should be that there is a linear relationship between at least one of the independent variables and the sales revenue (1 mark)

Notes:

No mark for stating the p-value, it can be read off from the output.

No mark for giving the decision (reject the null or not).

0.5 marks for stating that the model is significant.

2.2 It was found that the correlation between *AdSpend* and *Customers* is -0.57.

- a) What undesirable property does this value imply? (1)

Multicollinearity (1 mark)

- b) Give one consequence of the property that you identified in 2.2.a). (1)

Any one of the following can receive 1 mark (taken directly from the notes, no other answers to be accepted):

- a) Inflated standard errors
- b) Inflated p-values
- c) Large changes to parameter estimates when removing or adding a correlated independent variable from or to the model
- d) Increase in RSE when adding a correlated independent variable to the model

2.3 Why would the analyst be interested in including an interaction term in the model? (2)

The analyst must have suspected that the relationship between the sales revenue and the average price of the key product is affected by the store type (franchise or

independent) (1 mark), so that it would be necessary to modify the effect of the average price on the sales revenue for different store types (1 mark).

Notes:

For both marks, award only 0.5/1 if students do not refer to the specific situation and use generic x and y values.

- 2.4 Write down the regression equation for the full model when *Region* is East and *StoreType* is independent. Use the variable names in your equation and not nondescript x and y variables. (2)

$$\text{SalesRev} = (282.65619 + 46.79680) + 3.13544\text{AdSpend} + 2.05248\text{Customers} + 0.43694\text{StoreSize} - (1.01361 + 1.98533)\text{AvgPrice}$$

$$\text{SalesRev} = 329.45299 + 3.13544\text{AdSpend} + 2.05248\text{Customers} + 0.43694\text{StoreSize} - 2.99894\text{AvgPrice}$$

OR with rounded values:

$$\text{SalesRev} = 329.45 + 3.14\text{AdSpend} + 2.05\text{Customers} + 0.44\text{StoreSize} - 3\text{AvgPrice}$$

Award marks as follows:

2 marks if it's the correct final equation, otherwise:

0.5 marks for the correct final intercept at 329.45299 or 329.45

0.5 marks for the correct final slope coefficient for *AvgPrice* at 2.99894 or 3.00

0.5 marks for NOT including *Region* or setting all the *Region*-variables equal to 0

0.5 marks for variable names indicated

Subtract 0.5 if the final equation is not simplified for *StoreTypeIndependent* = 1.

Subtract 0.5 marks if the final equation is correct, but not simplified

- 2.5 Calculate the predicted sales revenue for a month where there is no advertising spend, there are 400 customers, the store size is 32 m², and the average price of the product is R200.00. The store is a franchise in the Northern parts of town. Show all your calculations. (2)

$$\text{SalesRev} = (282.65619 + 72.67649) + 2.05248\text{Customers} + 0.43694\text{StoreSize} - 1.01361\text{AvgPrice}$$

$$\text{SalesRev} = 355.33268 + 2.05248(4) + 0.43694(32) - 1.01361(200)$$

(1 mark for correct substitutions in terms of the units of measurement of the different variables)

The predicted sales revenue is $174.80268 \times 10\,000 = \text{R}1\,748\,026.80$. (1 mark)

Notes:

Subtract 0.5 if the answer is correct, but not translated to R... value.

Subtract 0.5 if the intermediary steps are rounded and they obtain R1 756 200 as final answer.

2.6 Interpret the estimated coefficient for *RegionNorth* in the R output. (2)

On average, the sales revenue is $(72.67649 \times 10\,000 =)$ R726 764.90 (0.5 marks) higher/increased (0.5 marks) for a store located in the Northern parts of the town (0.5 marks) compared to a store located in the Eastern parts of the town, while holding all else constant (0.5 marks).

Notes:

Do not mark specifically for *on average*.

Do not award marks if only 72.67649 is reported – the R... value needs to be calculated.

If the students do not explicitly state *...compared to a store in the Eastern parts...* they may still get the marks.

QUESTION 3 [9]

A retailer runs targeted campaigns and wants to predict whether a customer will make a purchase this month. They have data over 40 months on whether customer make a purchase, how much discount was offered to the customer, and via which marketing channel the customer was contacted.

A detailed description of the variables is listed below:

- Purchase Made (*Purchase*) — Yes (1) if the customer made a purchase this month, No (0) otherwise.
- Discount Offered (*Discount*) — percentage points (e.g., 0 to 100; not a fraction).
- Marketing Channel (*Channel*) — Email, social media (Social), independent internet search (Search).

The retailer has tasked a young statistician with choosing the correct model for this data and for coming up with an answer to their question. Consult the output generated from *R Studio* to answer the questions that follow.


```

Call:
glm(formula = Purchase ~ Discount + Channel, family = "binomial")

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -2.61496    1.13929  -2.295   0.02172 *
Discount       0.17979    0.05988   3.002   0.00268 **
ChannelSocial -0.44314    0.98168  -0.451   0.65170
ChannelSearch  0.36666    0.94730   0.387   0.69871
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 55.452  on 39  degrees of freedom
Residual deviance: 41.918  on 36  degrees of freedom
AIC: 49.918

Number of Fisher Scoring iterations: 4

```

- 3.1 Interpret the coefficient associated with the outcome that the marketing channel is social media, in terms of the odds. State your answer in terms of a percentage and indicate whether the odds increase or decrease. (2)

On average, there is a $(1 - e^{-0.44314}) \times 100 = 35.80\%$ decrease (1 mark) in the odds of a purchase taking place when the marketing channel is social media compared to when it is email (1 mark).

- 3.2 Calculate the predicted probability of a purchase when a discount of 18% is offered to a contact who was reached through independent internet search. Show all your calculations. (3)

$$\log\left(\frac{p_i}{1-p_i}\right) = (-2.61496 + 0.36666) + 0.17979 \text{Discount} = -2.2483 + 0.17979(18) = 0.98792 \text{ (2 marks)}$$

$$p_i = \frac{1}{1 + e^{-(0.98792)}} \approx 0.73 \text{ (1 mark)}$$

Notes:

Award 1 mark for correct substitution and 1 mark for the final predicted log-odds to make up the first 2 marks.

Can also award all three marks if students substituted in the log-odds equation into the natural e-function.

- 3.3 The following (incomplete) confusion matrix was created for a probability threshold of 0.4. It is known that Sensitivity = 0.85, and PPV = 0.7391. Use this information to determine the values of A and B. (2)

		Observed	
		1	0
Predicted	1	17	A
	0	B	14

$$PPV = \frac{TP}{TP+FP} = \frac{17}{17+FP} = 0.7391 \rightarrow FP = \frac{17-17(0.7391)}{0.7391} \approx 6 = A \text{ (1 mark)}$$

$$\text{Sensitivity} = \frac{TP}{TP+FN} = \frac{17}{17+FN} = 0.85 \rightarrow FN = \frac{17-17(0.85)}{0.85} = 3 = B \text{ (1 mark)}$$

3.4 The following output was generated from a model-building exercise that the statistician undertook to determine the best model to fit for this dataset.

```
Start: AIC=49.92
Purchase ~ Channel + Discount

          Df Deviance    AIC
- Channel  2   42.615 46.615
<none>          41.918 49.918
- Discount  1   55.308 61.308

Step: AIC=46.61
Purchase ~ Discount

          Df Deviance    AIC
<none>          42.615 46.615
- Discount  1   55.452 57.452

Call: glm(formula = Purchase ~ Discount, family = "binomial", data = data)

Coefficients:
(Intercept)      Discount
    -2.3849         0.1643

Degrees of Freedom: 39 Total (i.e. Null);  38 Residual
Null Deviance:      55.45
Residual Deviance: 42.61      AIC: 46.61
```

- a) Which model-building method was implemented to produce this output? (1)

Backwards elimination (1 mark)

- b) Why should we be cautious about including too many independent variables in our model fit? (1)

This can lead to the model fitting the dataset under consideration very well (precise) (0.5 marks) but can lead to the model not fitting unseen data well (biased) (0.5 marks).

Can also give 0.5 marks if they state something about 'overfitting'.

SECTION C: ANOVA QUESTIONS

[10]

QUESTION 1

[10]

A baking technologist is testing whether three baking-powder formulations affect the rise height of cupcakes. A single batch of standardised batter is prepared and divided into 15 identical portions. Each portion is randomly assigned to one of the three formulations: Formulation X, Formulation Y, Formulation Z. All cupcakes are baked in the same convection oven in one session and the pan positions are randomised.

Formulation	Heights (mm)	Sample SD
X	38.1, 41.0, 39.2, 36.9, 38.8	1.51
Y	34.0, 38.2, 36.9, 33.1, 35.9	2.08
Z	40.5, 42.0, 39.7, 41.2, 38.9	1.22

The overall mean height of cupcakes is 38.29 mm.

1. Identify the following based on the description and data: (2)
- a. Treatment factor(s)

Baking powder formulation

- b. Treatments

Three different baking powder formulations (X,Y,Z)

- c. Experimental unit

A portion of the batter / cupcakes / cupcake

- d. Number of replicates

Five

2. Explain what the fitted values of this model represent and give the actual numeric value for one of the treatments. (1)

The fitted values are mean cupcakes heights for each treatment. / sample means of treatments [0.5]

[0.5] for the correct mean:

Mean height for Formulation X = 38.8

Mean height for Formulation Y = 25.62

Mean height for Formulation Z = 40.46

3. For the treatment you chose above, calculate and interpret its effect. (1.5)

[0.5] for the correct effect calculate (mark with error from previous question)

Effect of Formulation X = $38.8 - 38.29 = 0.51$

Effect of Formulation Y = $35.62 - 38.29 = -2.67$

Effect of Formulation Z = $40.46 - 38.29 = 2.17$

No marks for doing the calculation the wrong way around.

The height of experimental units / cupcakes who received treatment [insert] were roughly [insert effect] higher/lower than average. [1] for correct interpretation. Increased or decreased height on average is not a sufficient answer. Stating something along the lines of the formulation X has an effect of 0.51 more than the overall mean is OK for full 1 mark.

4. An analyst notes that the sample standard deviations shown above are different, which means that the assumption of equal population variance has been violated. Do you agree with this statement? Motivate your answer. (1.5)

No / do not agree [0.5]. Just because they are different does not mean they do not have the same population variance [0.5], they can differ due to sampling variability [0.5]

5. Using the data above, the following ANOVA table was produced in R.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
formulation	2	60.49	30.245	11.21	0.0018
Residuals	12	32.38	2.698		

- a. Write (in words) the null and alternative hypotheses being tested here in the context of the experiment. (1)

H0: The mean height of cupcakes is the same across the formulation types. / All the treatment effects are zero.

H1: At least one formulation type resulted in a different mean cupcake height. / one treatment effect is not zero.

No marks for saying all the means are different or effects are all different etc.

Half mark for a vague set of hypothesis statements: H0, all are the same vs h1: Atleast one is different.

- b. What does the mean square associated with the variable "formulation" represent?(1)

The average between group variability / It measures the variability of treatment means around the overall / amount of variation explained by the treatments [1]

Variation between formulations/treatments is worth [0.5]

- c. Conclude the hypothesis test using the p-value approach and a significance level of 1%. (2)

Since the p-value is smaller than the significance level of 1% [0.5], there is enough evidence to reject the null hypothesis [0.5] and conclude that at least one treatment mean is different [1].

Mark with error. First part wrong, but the rest is correct, give the appropriate marks. Using 5% is wrong. Saying that the p-value is sufficiently small is good enough for [0.5].

STA2020F: Important Formulae

SIMPLE LINEAR REGRESSION

Sum of squares:

$$SS_{reg} = SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

$$SS_{resid} = SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$SS_{tot} = SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

Measures of association:

$$r_{xy} = \frac{SS_{xy}}{\sqrt{SS_x SS_y}}$$

Standard error of the regression slope:

$$s_{\beta_1} = \frac{s_{\varepsilon}}{\sqrt{SS_x}}, \text{ where } RSE = s_{\varepsilon}$$

$(1 - \alpha)\%$ confidence interval for β_j :

$$\hat{\beta}_j \pm t_{\frac{\alpha}{2}, n-p-1} s_{\beta_j}$$

MULTIPLE REGRESSION

Multiple regression model (for the sake of defining n and p):

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i,1} + \hat{\beta}_2 x_{i,2} + \cdots + \hat{\beta}_p x_{i,p} \text{ for } i = 1, 2, 3, \dots, n$$

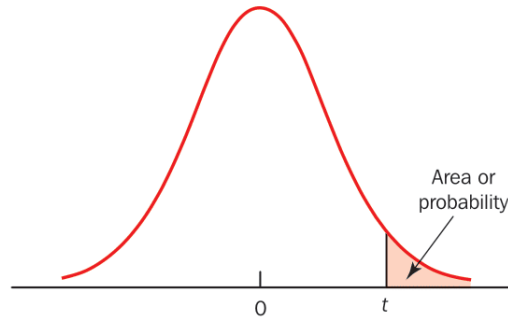
LOGISTIC REGRESSION

logit: $\log\left(\frac{p}{1-p}\right)$

Inverse logit: $p = \frac{1}{1 + \exp^{-(\beta_0 + \beta_1 x + \cdots)}}$

$(1 - \alpha)\%$ confidence interval for β_j :

$$\hat{\beta}_j \pm Z_{\frac{\alpha}{2}} s_{\beta_j}$$

TABLE 2 *t* distribution

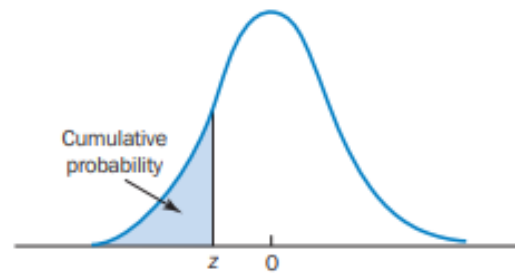
Entries in the table give *t* values for an area or probability in the upper tail of the *t* distribution. For example, with ten degrees of freedom and a .05 area in the upper tail, $t_{.05} = 1.812$.

Degrees of freedom	Area in upper tail					
	.20	.10	.05	.025	.01	.005
1	1.376	3.078	6.314	12.706	31.821	63.656
2	1.061	1.886	2.920	4.303	6.965	9.925
3	.978	1.638	2.353	3.182	4.541	5.841
4	.941	1.533	2.132	2.776	3.747	4.604
5	.920	1.476	2.015	2.571	3.365	4.032
6	.906	1.440	1.943	2.447	3.143	3.707
7	.896	1.415	1.895	2.365	2.998	3.499
8	.889	1.397	1.860	2.306	2.896	3.355
9	.883	1.383	1.833	2.262	2.821	3.250
10	.879	1.372	1.812	2.228	2.764	3.169
11	.876	1.363	1.796	2.201	2.718	3.106
12	.873	1.356	1.782	2.179	2.681	3.055
13	.870	1.350	1.771	2.160	2.650	3.012
14	.868	1.345	1.761	2.145	2.624	2.977
15	.866	1.341	1.753	2.131	2.602	2.947
16	.865	1.337	1.746	2.120	2.583	2.921
17	.863	1.333	1.740	2.110	2.567	2.898
18	.862	1.330	1.734	2.101	2.552	2.878
19	.861	1.328	1.729	2.093	2.539	2.861
20	.860	1.325	1.725	2.086	2.528	2.845
21	.859	1.323	1.721	2.080	2.518	2.831
22	.858	1.321	1.717	2.074	2.508	2.819
23	.858	1.319	1.714	2.069	2.500	2.807
24	.857	1.318	1.711	2.064	2.492	2.797
25	.856	1.316	1.708	2.060	2.485	2.787
26	.856	1.315	1.706	2.056	2.479	2.779
27	.855	1.314	1.703	2.052	2.473	2.771
28	.855	1.313	1.701	2.048	2.467	2.763
29	.854	1.311	1.699	2.045	2.462	2.756

TABLE 2 (Continued)

Degrees of freedom	Area in upper tail					
	.20	.10	.05	.025	.01	.005
30	.854	1.310	1.697	2.042	2.457	2.750
31	.853	1.309	1.696	2.040	2.453	2.744
32	.853	1.309	1.694	2.037	2.449	2.738
33	.853	1.308	1.692	2.035	2.445	2.733
34	.852	1.307	1.691	2.032	2.441	2.728
35	.852	1.306	1.690	2.030	2.438	2.724
36	.852	1.306	1.688	2.028	2.434	2.719
37	.851	1.305	1.687	2.026	2.431	2.715
38	.851	1.304	1.686	2.024	2.429	2.712
39	.851	1.304	1.685	2.023	2.426	2.708
40	.851	1.303	1.684	2.021	2.423	2.704
41	.850	1.303	1.683	2.020	2.421	2.701
42	.850	1.302	1.682	2.018	2.418	2.698
43	.850	1.302	1.681	2.017	2.416	2.695
44	.850	1.301	1.680	2.015	2.414	2.692
45	.850	1.301	1.679	2.014	2.412	2.690
46	.850	1.300	1.679	2.013	2.410	2.687
47	.849	1.300	1.678	2.012	2.408	2.685
48	.849	1.299	1.677	2.011	2.407	2.682
49	.849	1.299	1.677	2.010	2.405	2.680
50	.849	1.299	1.676	2.009	2.403	2.678
51	.849	1.298	1.675	2.008	2.402	2.676
52	.849	1.298	1.675	2.007	2.400	2.674
53	.848	1.298	1.674	2.006	2.399	2.672
54	.848	1.297	1.674	2.005	2.397	2.670
55	.848	1.297	1.673	2.004	2.396	2.668
56	.848	1.297	1.673	2.003	2.395	2.667
57	.848	1.297	1.672	2.002	2.394	2.665
58	.848	1.296	1.672	2.002	2.392	2.663
59	.848	1.296	1.671	2.001	2.391	2.662
60	.848	1.296	1.671	2.000	2.390	2.660
61	.848	1.296	1.670	2.000	2.389	2.659
62	.847	1.295	1.670	1.999	2.388	2.657
63	.847	1.295	1.669	1.998	2.387	2.656
64	.847	1.295	1.669	1.998	2.386	2.655
65	.847	1.295	1.669	1.997	2.385	2.654
66	.847	1.295	1.668	1.997	2.384	2.652
67	.847	1.294	1.668	1.996	2.383	2.651
68	.847	1.294	1.668	1.995	2.382	2.650
69	.847	1.294	1.667	1.995	2.382	2.649

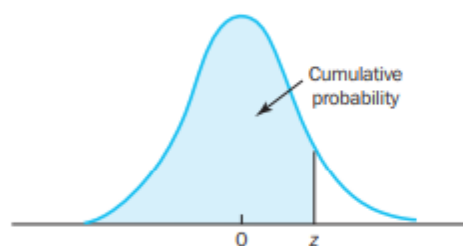
(continued)

TABLE 1 Cumulative Probabilities for the Standard Normal Distribution

Entries in the table give the area under the curve to the left of the z value. For example, for $z = -.85$, the cumulative probability is .1977.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

TABLE 1 (Continued)



Entries in the table give the area under the curve to the left of the z value. For example, for $z = 1.25$, the cumulative probability is .8944.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9913
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9986	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990